

Owed lemmas for the rank-two atomic residue proof

Tavis Rudd

August 2026

Purpose. The argument transcribed in `2026-08-15-c912-atom-residue-proof.tex` uses, at its first and most load-bearing step, horizontality of the Poincaré pairing for the A -model connection. Its independent verification (`2026-08-15-c912-atom-residue-proof-verification.md`) found that this is *not* an available import: Katzarkov–Kontsevich–Pantev–Yu state horizontality only after assuming convergence and passing to the complex-analytic bundle, whereas the A -model F -bundle of a projective variety is non-archimedean analytic. The step is therefore ours to prove. This note proves it, over the Novikov ring and with no convergence hypothesis, and lists what else that verification recorded as owed.

1 Setting

Let T be a smooth projective variety over $\mathbf{C}$ of dimension n , and let Λ be the completed Novikov ring of its numerical curve lattice. All statements below are identities in Λ -modules; no convergence of any Gromov–Witten generating function is used, and no base change to a complex-analytic bundle is performed.

Write $H = H^{\text{ev}}(T) \otimes \Lambda$ for the even part, $(\cdot, \cdot)$ for the Poincaré pairing extended Λ -bilinearly, $\star_\tau$ for the genus-zero quantum product at a bulk parameter τ , and

$$\text{Gr}(\phi) = \frac{1}{2}(\deg \phi - n)\phi \quad \text{for homogeneous } \phi,$$

for the grading operator. The connection is

$$\nabla_u \partial_u = u \partial_u - u^{-1} \mathcal{E} \star_\tau + \text{Gr}, \quad \nabla_\xi = \xi + u^{-1} C_\xi, \quad C_\xi = \xi \star_\tau, \quad (1)$$

so that horizontal sections satisfy

$$u \partial_u y = A(u)y, \quad A(u) = u^{-1}U - \text{Gr}, \quad U := \mathcal{E} \star_\tau; \quad \xi y = B_\xi(u)y, \quad B_\xi(u) = -u^{-1}C_\xi. \quad (2)$$

The pairing is taken in the usual twisted form: for sections $y, \tilde{y}$ one considers $(y(u), \tilde{y}(-u))$.

2 The two adjointness identities

Lemma 2.1 (Frobenius, over Λ). *For all $a, b, c \in H$ and every bulk parameter τ ,*

$$(a \star_\tau b, c) = (a, b \star_\tau c).$$

In particular every quantum multiplication operator, including $U = \mathcal{E} \star_\tau$ and each $C_\xi = \xi \star_\tau$, is self-adjoint for the Poincaré pairing.

Remark 2.2 (The convention this depends on). That U is a quantum multiplication operator is a convention-sensitive point and should be cited rather than assumed: in several presentations the Euler operator on a quantum connection carries an additional derivation term, and if it did here the lemma would not apply to it. It does not. In Katzarkov–Kontsevich–Pantev–Yu, (3.31) makes $\mathcal{E}_\gamma$ a cohomology class and (3.30) makes the operator $\mathcal{O}$ -linear multiplication by it, with no derivation term.

Proof. By definition the structure constants of $\star_\tau$ are

$$(a \star_\tau b, c) = \sum_d \sum_{m \geq 0} \frac{Q^d}{m!} \langle a, b, c, \tau, \dots, \tau \rangle_{0, m+3, d},$$

a formal sum over effective numerical classes d with coefficients the genus-zero Gromov–Witten invariants. Those invariants are defined by integration against the virtual class of a moduli space of stable maps with *unordered* marked-point labels up to the specified insertions, hence are symmetric under permuting their arguments; on the even part no Koszul signs arise. The displayed expression is therefore totally symmetric in a, b, c , and comparing it with the same expression for $(a, b \star_\tau c)$ gives the claim.

Two remarks on what this does and does not use. The identity is an identity of formal series: it holds coefficientwise in Q^d and in the bulk parameter, so it is a statement in Λ and requires no convergence. And it is exactly the total symmetry of the third derivatives of the genus-zero potential, so it is the Frobenius property in its usual form, not a strengthening of it. $\square$

Lemma 2.3 (The grading operator is anti-self-adjoint). *For all $a, b \in H$, $(\text{Gr}a, b) + (a, \text{Gr}b) = 0$.*

Proof. Both sides are Λ -bilinear, so it suffices to check on homogeneous classes. The Poincaré pairing of $\phi \in H^k$ with $\psi \in H^\ell$ vanishes unless $k + \ell = 2n$. In that case

$$(\text{Gr}\phi, \psi) + (\phi, \text{Gr}\psi) = \left[\frac{1}{2}(k - n) + \frac{1}{2}(\ell - n) \right] (\phi, \psi) = \frac{1}{2}[(k + \ell) - 2n] (\phi, \psi) = 0.$$

This is a statement about the cohomological grading and Poincaré duality alone; the Novikov variables play no role, and Gr is Λ -linear because it acts on the coefficient ring trivially. $\square$

Remark 2.4. Lemma 2.3 is where the normalization of Gr by $-n/2$ earns its keep: with any other additive constant the operator would fail to be anti-self-adjoint, and horizontality below would acquire a scalar defect.

3 Horizontality

Theorem 3.1 (The Poincaré pairing is horizontal, over Λ). *Let $y, \tilde{y}$ be horizontal sections of (1). Then*

$$(y(u), \tilde{y}(-u))$$

is constant in u and in every base direction. Equivalently, with P_0 the Gram matrix of the Poincaré pairing in a fixed basis,

$$A(u)^T P_0 + P_0 A(-u) = 0, \quad B_\xi(u)^T P_0 + P_0 B_\xi(-u) = 0. \quad (3)$$

Proof. Put $f(u) = y(u)^T P_0 \tilde{y}(-u)$. Substituting $w = -u$ shows $u\partial_u$ is invariant under $u \mapsto -u$, so $(u\partial_u)(\tilde{y}(-u)) = A(-u)\tilde{y}(-u)$, and therefore

$$u\partial_u f = y^T [A(u)^T P_0 + P_0 A(-u)] \tilde{y}(-u).$$

Now compute the bracket from (2):

$$(u^{-1}U - \text{Gr})^T P_0 + P_0(-u^{-1}U - \text{Gr}) = u^{-1}(U^T P_0 - P_0 U) - (\text{Gr}^T P_0 + P_0 \text{Gr}).$$

The first bracket vanishes by Lemma 2.1, since U is a quantum multiplication operator, and the second by Lemma 2.3. Hence $u\partial_u f = 0$.

For a base direction, $B_\xi(u) = -u^{-1}C_\xi$ has no Gr term and is odd in u , so

$$B_\xi(u)^T P_0 + P_0 B_\xi(-u) = -u^{-1}(C_\xi^T P_0 - P_0 C_\xi) = 0$$

by Lemma 2.1 again, applied to $C_\xi = \xi \star_\tau$. So f is constant in every base direction as well. $\square$

Corollary 3.2 (What transfers to the atomic residue proof). *Equation (7) of 2026-08-15-c912-atom-residue holds on the even bundle in the global cohomology frame, with the constant Gram matrix P_0 . Its consequences (8), (10) and (11) follow in that frame.*

Proof. Immediate from Theorem 3.1: with $P(u) = P_0$ constant the $u\partial_u P$ term of (7) drops and the remainder is (3). The u^{-1} coefficient is $N^T P_0 = P_0 N$, which is (8), and the sandwich argument of the target then gives $A_0 L \subset L$. $\square$

Remark 3.3 (A correction: the pairing is *not* constant on a spectral factor, and the corollary must not claim it is). An earlier version of this corollary asserted additionally that (10) holds “with $P_1 = 0$ ”, on the ground that the pairing is constant. That is **false as a general statement** and the error is worth recording, since it is the kind that hides in bookkeeping.

Theorem 3.1 is proved in the global cohomology frame, where the Gram matrix is the constant P_0 . The target’s equation (7) is applied in the even *spectral-factor* frame of a single atom, which is reached from the global frame by a u -dependent gauge F . In that frame the pairing is

$$P(u) = F(u)^T P_0 F(-u) = P_0|_{\text{block}} - u^2 \sum F_1^T P_0 F_1 + O(u^3),$$

which is not constant; P_1 vanishes only in a particular block-off-diagonal Sylvester gauge, so “ $P_1 = 0$ ” is gauge-dependent bookkeeping and not a fact about the pairing.

Nothing downstream is affected. The target’s own sandwich argument derives (8) and $A_0 L \subset L$ with P_1 left arbitrary, which is exactly why it was written that way; the simplification claimed here was never needed. What the corollary may be cited for is the statement above, in the global frame, and no more.

Remark 3.4 (What was actually at issue). The verification’s concern was not that horizontality is false but that the source establishes it only after assuming convergence and passing to the complex-analytic bundle, while the object the proof works with is non-archimedean. Theorem 3.1 removes that dependence: both inputs are algebraic identities in Λ -modules, one the total symmetry of genus-zero Gromov–Witten invariants and the other a statement about the cohomological grading. Neither refers to a topology on Λ . In particular nothing here touches the Euler-pairing comparison or the Serre automorphism, which are the objects the source does defer.

4 Still owed

The same verification recorded four further unstated inputs. They are bookkeeping rather than substance, but none is written:

- (i) that the even part is a sub- F -bundle, so that restricting the whole construction to it is legitimate — note that Corollary 3.2 does *not* discharge this, and an earlier version of that corollary silently consumed it, which was circular;
- (i') the transfer of Theorem 3.1 from the global frame to a single even spectral factor, carrying a u -dependent $P(u)$ as in Remark 3.3, together with spectral orthogonality and non-degeneracy of the restricted pairing on that factor. This is the genuine residue of the present note and is what the target's (7) actually assumes;
- (ii) uniqueness and gluing of the spectral splitting;
- (iii) the identity principle on irreducible k -analytic spaces, which is what the crossing-point argument uses when it propagates constancy from a nonempty open subset;
- (iv) descent of the invariant through the blowup, projective-bundle and disjoint-union equivalences, and not only through the two equivalences in the source's own definition.

Separately, and not bookkeeping: the parity ranks $(2, 10)$ rest on the source's assertion that the zero-eigenvalue packet does not split further at the relevant point, whose cited support carries no proof and covers only a restricted case. That is the one place where the argument is genuinely conditional, and it is a finite splitting statement about a single packet rather than an invariance statement about arbitrary centres.